

THE MIND GENESIS ARCHITECTURE (TMGA)

REGULATORY MAPPING & REFERENCE MATRIX

Conceptual Cross-Standard Mapping of TMGA's 33 Dimensions, NHCP, Cryptographic Verification, Shadow Auditing, and Circuit Breaker Mechanisms to NIST AI RMF 1.0 and ISO/IEC 42001:2023

Executive Summary & Scope

The Mind Genesis Architecture (TMGA) is a proposed 33-dimensional reference architecture for studying how cognitive capability, verification, contextual interpretation, operational restraint, and governance may interact within shared intelligence ecosystems.

This document provides a conceptual cross-standard mapping between selected TMGA architectural mechanisms and relevant functions or control areas within the NIST AI Risk Management Framework (AI RMF 1.0) and ISO/IEC 42001:2023.

The mapping is intended as an architectural reference and research aid. It does not constitute certification, conformity assessment, regulatory approval, or evidence that TMGA satisfies the requirements of NIST AI RMF, ISO/IEC 42001, or any other regulatory or technical standard.

The relationships identified below indicate areas in which TMGA mechanisms may support, inform, or be evaluated against corresponding risk-management and governance objectives. Actual conformity would depend on deployment-specific implementation, organizational processes, independent assessment, empirical validation, and applicable legal or regulatory requirements.

The 33 TMGA dimensions themselves are a proposed architectural taxonomy rather than empirically established components of cognition. Similarly, the engineering mechanisms described here—including NHCP, cryptographic verification, asynchronous auditing, hardware-assisted containment, and the Circuit Breaker Protocol—should be understood as reference designs unless explicitly identified as implemented and evaluated within the accompanying simulation environment.

1. NIST AI RMF 1.0 Core Functions — Conceptual Mapping

NIST AI RMF 1.0 organizes AI risk management around four core functions: GOVERN, MAP, MEASURE, and MANAGE. The following table identifies proposed correspondences between these functions and selected TMGA mechanisms.

NIST AI RMF Core	Risk-Management Objective	Proposed TMGA Mapping	Illustrative Architectural Mechanism
GOVERN	Policies, processes, accountability structures, and organizational oversight for AI risk	Sphere III: AWI; AWI Systems (3.9); AWI Safe (3.6); governance constraints	NHCP-based architectural arbitration, governance-state representation, auditability, and authorization constraints
MAP	Context identification, capability characterization, task boundaries, affected stakeholders, and hazard identification	Sphere I & III; AWI Mind (3.1); AGI/ASI Fold (1.4/2.4); AWI Void (3.7)	Context representation, trade-off modeling, state-space analysis, and explicit representation of uncertainty
MEASURE	Quantitative and qualitative evaluation, monitoring, drift assessment, and verification	Sphere I, II & III; Shadow Auditing; AWI Sentry (3.8); AGI/ASI Verify (1.10/2.11)	Drift measurement, asynchronous monitoring, formal predicate verification, and illustrative cryptographic proof relations
MANAGE	Risk response, operational constraints, escalation, containment, and incident response	CBP; AWI Void (3.7); bounded degradation; proposed hardware-assisted containment	Temporary containment states, suspension rules, escalation pathways, bounded degradation, and deployment-specific enforcement mechanisms

Interpretive note: These correspondences represent architectural relationships rather than demonstrated compliance with individual NIST AI RMF outcomes.

2. ISO/IEC 42001:2023 — Reference Alignment

ISO/IEC 42001 establishes an Artificial Intelligence Management System (AIMS). The table below identifies areas in which proposed TMGA mechanisms may provide technical or architectural support for related management-system objectives.

ISO/IEC 42001 Area	General Objective	Proposed TMGA Feature	Illustrative Evaluation or Evidence
Risk identification and treatment	Identify, assess, document, and respond to AI-related risks	AWI Void (3.7), uncertainty representation,	Evaluation of modeled confidence and uncertainty conditions; suspension or

ISO/IEC 42001 Area	General Objective	Proposed TMGA Feature	Illustrative Evaluation or Evidence
		restraint mechanisms	escalation under specified thresholds
AI system impact assessment	Evaluate potential societal, ethical, environmental, and stakeholder impacts	AWI Fold (3.5), contextual evaluation, stakeholder-oriented governance	Structured trade-off analysis, counterfactual assessment, and documented impact reasoning
Operational control	Maintain operation within defined organizational and technical constraints	Cryptographic verification pipeline, AGI/ASI Verify	Verification of explicitly specified predicates and constraints; illustrative ZK proof relations
Monitoring and internal audit	Evaluate performance, behavior, and governance processes over time	Asynchronous Shadow Auditing, AWI Sentry (3.8)	State snapshots, drift monitoring, append-only audit records, and deployment-dependent auditing intervals
Nonconformity and corrective action	Respond to failures, deviations, or unacceptable operating conditions	Circuit Breaker Protocol (CBP), escalation and suspension mechanisms	Temporary containment, recorded escalation, audit trails, and proposed deployment-specific enforcement

Important: TMGA does not by itself establish ISO/IEC 42001 conformity. Organizational governance, documented management processes, independent assessment, and implementation-specific evidence would remain necessary.

3. Master 33-Dimension Reference Mapping Matrix

The following matrix maps each proposed TMGA dimension to potentially relevant risk-management and governance areas. These mappings are exploratory architectural correspondences and should not be interpreted as one-to-one regulatory controls.

Sphere & Dimension	Primary Architectural Role	NIST AI RMF Reference Area	ISO/IEC 42001 Reference Area	Potential Audit / Evaluation Artifact
1.1 AGI Mind	Cross-domain integration	MAP / MANAGE	System scope and context	Context classification trace
1.2 AGI Deep	Conceptual synthesis and abstraction	MAP / MEASURE	Impact assessment	Causal abstraction log
1.3 AGI	Executive functioning	GOVERN /	Governance	Executive task-

Sphere & Dimension	Primary Architectural Role	NIST AI RMF Reference Area	ISO/IEC 42001 Reference Area	Potential Audit / Evaluation Artifact
Cortex	and memory coordination	MANAGE	structure	allocation trace
1.4 AGI Fold	Non-linear state-space modeling	MAP / MEASURE	Risk treatment	High-dimensional trade-off matrix
1.5 AGI Synapse	Modular signal routing	MANAGE / MAP	Technical resources	Agent interconnect trace
1.6 AGI Pulse	Persistent state tracking	MEASURE / MANAGE	Monitoring	State telemetry register
1.7 AGI Omni	Multimodal stream synthesis	MAP / MEASURE	Data governance	Multimodal processing log
1.8 AGI Guard	Sandboxed execution boundaries	MANAGE / GOVERN	Boundary controls	Sandbox isolation record
1.9 AGI Sentry	Behavioral auditing and drift detection	MEASURE / MANAGE	Internal audit	Behavioral drift alert
1.10 AGI Verify	Formal property checking	MEASURE / GOVERN	Verification	Formal verification record
1.11 AGI Void	Epistemic uncertainty modeling	MAP / MEASURE	Risk identification	Uncertainty boundary record
1.12 AGI Matter	Embodied physical interaction	MANAGE / MAP	Physical-system controls	Actuator-boundary telemetry
1.13 AGI Wake	Recursive metacognitive assessment	GOVERN / MEASURE	Management review	Self-assessment diagnostic
2.1 ASI Mind	Integration of high-scale cognitive state	MEASURE / MAP	Compute infrastructure	High-scale execution log
2.2 ASI Wake	Scale-state tracking	MEASURE / MANAGE	Monitoring	High-throughput state capture
2.3 ASI Deep	Large-scale hypothesis generation	MAP / MEASURE	Data synthesis	Discovery audit trail
2.4 ASI Fold	Systemic complex optimization	MANAGE / MAP	Systemic risk treatment	Multi-variable optimization trace
2.5 ASI Thread	Distributed synchronization	MANAGE / GOVERN	Network resilience	Consensus latency log
2.6 ASI Bloom	Accelerated domain expansion	MAP / MANAGE	Capability monitoring	Capability-boundary record
2.7 ASI Dark	Opacity management and interpretability	MEASURE / GOVERN	Transparency controls	Interpretability / proof artifact
2.8 ASI Guard	Cryptographic isolation boundaries	MANAGE / GOVERN	Security controls	Cryptographic barrier record
2.9 ASI	High-precision physical	MANAGE /	Physical-system	Hardware-interlock

Sphere & Dimension	Primary Architectural Role	NIST AI RMF Reference Area	ISO/IEC 42001 Reference Area	Potential Audit / Evaluation Artifact
Matter	interfaces	MAP	controls	status
2.10 ASI Void	Unmodeled systemic-risk representation	MAP / MANAGE	Incident management	Systemic fragility alert
2.11 ASI Verify	Formal verification at scale	MEASURE / GOVERN	Verification	Cryptographic or formal proof artifact
3.1 AWI Mind	Contextual integration and semantic interpretation	MAP / GOVERN	Governance context	Semantic intent classification
3.2 AWI Core	Neural-symbolic constraint integration	GOVERN / MEASURE	Rule consistency	Constraint-evaluation record
3.3 AWI Hub	Multi-agent and human-AI coordination	GOVERN / MANAGE	Stakeholder coordination	Human-governance interaction log
3.4 AWI Deep	Counterfactual consequence evaluation	MAP / MEASURE	Impact assessment	Counterfactual analysis report
3.5 AWI Fold	Societal and ecological trade-off analysis	GOVERN / MAP	Societal impact assessment	Structured trade-off matrix
3.6 AWI Safe	Normative and constitutional constraint representation	GOVERN / MANAGE	Governance controls	Constraint-check record
3.7 AWI Void	Epistemic restraint under uncertainty	MANAGE / MAP	Operational restraint	Restraint trigger and confidence record
3.8 AWI Sentry	Governance-baseline drift monitoring	MEASURE / GOVERN	Internal monitoring	Drift report
3.9 AWI Systems	Institutional and governance translation	GOVERN / MANAGE	Management-system alignment	Auditable governance record

Mapping limitation

The entries above identify plausible relationships between proposed TMGA functions and risk-management objectives. They do not establish that a TMGA dimension independently fulfills a regulatory requirement. In deployed systems, evidence would need to be produced at the organizational, procedural, software, hardware, and governance levels appropriate to the relevant standard.

4. Emergency Controls & Proposed Hardware-Assisted Enforcement

TMGA distinguishes between mechanisms exercised within the reference simulation and deployment-oriented mechanisms proposed for future implementation.

The reference simulations evaluate governance predicates, suspension conditions, drift monitoring, and modeled containment states. They do **not** physically instantiate Non-Maskable Interrupts, escrow buffers, actuator isolation, or hardware roots of trust.

Control Mechanism	Proposed Activation Condition	Reference Architectural Response	Current Evidence Status
Circuit Breaker Protocol (CBP)	Persistent drift above a specified threshold across consecutive auditing cycles under the applicable predicate	Temporary containment state with bounded scope, auditability, and suspension mechanisms	Implemented at reference-simulation level; not validated as a production containment system
Hardware-Assisted Interruption & Escrow	Deployment-specific safety predicate, guardrail failure, or unacceptable uncertainty	Proposed processor-level interruption, command buffering, and actuator isolation	Architectural proposal; not physically implemented or experimentally validated in the reference simulation
Bounded Operational Degradation	Computational stress or resource pressure	Risk-based reduction of selected functions while retaining specified governance and verification functions	Partially represented in the reference simulation; deployment behavior remains implementation-dependent

CBP and non-hierarchy

The Circuit Breaker Protocol should be interpreted as an exceptional containment state rather than a superior decision-making layer. Its authority is conditional on defined activation criteria and is intended to remain bounded in scope and duration. No component is assigned permanent unilateral jurisdiction over the architecture.

Hardware timing parameters

Any timing values used in architectural examples are illustrative engineering parameters rather than empirically established TMGA performance guarantees. Real deployment requirements would depend on processor architecture, actuator characteristics, communication latency, operational domain, and safety case.

5. Institutional Evaluation & Implementation Research Roadmap

The following steps describe possible areas for future implementation and evaluation. They are not a certification checklist.

1. Governance and Context Evaluation

Assess whether deployment-specific governance constraints, authorization rules, human oversight responsibilities, and escalation pathways are explicitly defined and auditable.

Evaluate whether relevant TMGA contextual and governance functions can be translated into measurable implementation requirements.

2. Runtime Monitoring and Verification

Evaluate whether drift-monitoring signals can be operationally defined and calibrated.

Test whether audit records remain complete and tamper-evident under realistic operating conditions.

Benchmark cryptographic or formal verification mechanisms against deployment-specific latency and computational requirements.

3. Crisis and Containment Evaluation

Implement containment mechanisms on representative hardware before making claims about physical interruption or actuator isolation.

Measure interruption latency, command-buffer behavior, actuator response, recovery procedures, and failure modes.

Evaluate CBP activation criteria against independently defined ground truth to distinguish appropriate containment from unnecessary or missed activation.

4. Independent Governance Assessment

Evaluate the proposed mapping against actual organizational obligations, applicable regulation, NIST AI RMF implementation profiles, and ISO/IEC 42001 requirements.

Where conformity or certification is sought, use qualified independent assessment rather than architectural mapping alone.

6. Interpretation and Limitations

This matrix should be read as a **research-oriented architectural crosswalk**.

It demonstrates that TMGA contains proposed functions that can be meaningfully compared with established AI risk-management and management-system frameworks. It does not demonstrate that:

- TMGA is NIST- or ISO-certified;
- the 33-dimensional taxonomy has been empirically validated;
- individual dimensions satisfy regulatory requirements by themselves;
- the complete architecture has undergone formal verification;
- proposed hardware containment mechanisms have been implemented;
- cryptographic verification establishes ethical correctness;
- simulation outcomes establish real-world safety or robustness; or
- implementation of TMGA would automatically produce regulatory conformity.

The value of this mapping lies in identifying testable relationships between architectural mechanisms and governance objectives. Whether those relationships remain effective in deployed systems is an empirical and institutional question requiring further evaluation.

Document Status

Supporting research documentation for The Mind Genesis Architecture (TMGA), 2026.

This document accompanies:

The Mind Genesis Architecture: A Multidimensional Model for Shared Intelligence Ecosystems.

Project repository:

<https://github.com/ahmett76/the-mind-genesis-architecture>